

Luis Alfredo Cuamatzi Flores

AI Engineer / Software Developer

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SUMMARY

AI Engineer and Software Developer with experience in mobile applications, backend services, and AI model integration. I have worked on real-world projects combining Android, APIs, automation, data processing, CRM, computer vision, and product-oriented interfaces to connect users, data, and business processes.

EXPERIENCE

Software & AI Developer

Jun 2025 – Present

Net2Alliance

Mexico City, Mexico

- Full-stack development and AI integration for telecom and CRM platforms, working with FastAPI, Python, PHP, JavaScript, WebSockets, and AWS SageMaker.
- Built 2 modules and 2 endpoints for sentiment analysis and automated PDF summary generation from call transcripts using AWS SageMaker.
- Developed a WebSocket-based WhatsApp messaging client with real-time notifications, multimedia support, and agent transfer workflows.
- Implemented 4 CRM modules with 4 endpoints, CRUD operations, access control for 3 user roles, audit logging, and administrative reporting.
- Contributed to UI/UX redesigns with Figma and task tracking using Jira, GitHub, and Kanban boards.

AI & Backend Engineer – Graduation Project / R&D

Jan 2025 – Dec 2025

L4L – 3D Garment Reconstruction for Virtual Try-On

Tlaxcala, Mexico

- Designed a client-server architecture for Virtual Try-On, integrating an Android app built with Jetpack Compose, a Linux inference server, and a Windows server for physical simulation and FBX-GLB conversion.
- Developed a FastAPI orchestrator to coordinate image capture, avatar generation and confirmation, garment reconstruction, physical simulation, and delivery of 3D models in GLB format.
- Integrated a single-image 2D body pose and shape estimation model (SMPLer-X), adapting its output for compatibility with the simulation pipeline.
- Adapted a two-level Transformer model for sewing pattern reconstruction from a 2D image (Sewformer), evaluating pruning, PTQ, and QAT variants with geometric and connectivity metrics.
- Compressed the final model with a 5.43x compression ratio and a 38.75% latency reduction, reaching 32.26 MB and 165.63 ms CPU inference time.

Android Developer – Social Service

Oct 2024 – Aug 2025

Instituto Politécnico Nacional – Institutional Tutoring Program

Tlaxcala, Mexico

- Implemented QR-based attendance scanning with real-time validation, duplicate detection, and active session control.
- Built a 3-role user system (Admin, Peer Tutor, and Advisor) with separate panels based on permissions.
- Built a weekly schedule generator with overlap detection for classrooms and tutors.
- Implemented offline-first synchronization with Firestore and CSV export for institutional reporting.

EDUCATION

B.Eng. in Artificial Intelligence Engineering

Aug 2021 – Jan 2026

Instituto Politécnico Nacional

Tlaxcala, Mexico

Leadership Formation Program

Aug 2024

Queen Mary University of London

London, UK

PROJECTS

Lexfy <i>Android + AI Project</i> - Android app with OCR and AI image generation through a chat interface. - Android client communicating via HTTP with a local Python/Flask backend running EasyOCR, GOT-OCR2_0, and Together AI. Tech: Kotlin, Jetpack Compose, Python, Flask, Firebase, EasyOCR, GOT-OCR2_0, Transformers.	Dec 2024 GitHub
SmartEMG Vision <i>Android prototype for assistive interaction</i> - Mobile prototype combining EMG signal classification and object detection to suggest context-aware actions. - Android client with Jetpack Compose and CameraX sending data via HTTP to local Python servers running YOLOv8 and TensorFlow/Keras. Tech: Kotlin, Jetpack Compose, CameraX, Python, Flask, TensorFlow, Keras, YOLOv8.	Nov 2024 GitHub
SmartCrops <i>Mobile prototype for greenhouse monitoring</i> - Prototype for crop monitoring, greenhouse variable simulation, and AI-based chatbot assistance (TomaBot, powered by Gemini). - Single-activity Jetpack Compose app with CameraX, OkHttp, and Coil. Tech: Kotlin, Jetpack Compose, Gemini API, CameraX, OkHttp, Coil.	Aug 2024 GitHub
CommuniSync <i>Android community platform</i> - Built a native Android app for neighborhood engagement, including civic issue reports, polls, forums, a local directory, notifications, user profiles, and guest access. - Integrated Firebase Authentication and Firestore for email, Google, and guest sign-in, data storage, and real-time synchronization. Tech: Kotlin, Jetpack Compose, Material 3, Firebase Auth, Firestore, Coil.	May 2024 GitHub
EduStream API <i>Web-based educational content management platform</i> - Web platform for uploading, categorizing, and streaming educational videos by subject. - MVC backend with multimedia file management and synchronization between physical storage and the database. Tech: Node.js, Express, MySQL, Handlebars, Bootstrap, Multer.	Jun 2023 GitHub

TECHNICAL SKILLS

Languages: Kotlin, Python, JavaScript, TypeScript, PHP, SQL
Mobile: Native Android, Jetpack Compose, Android Views/XML, Material 3, CameraX, Room, WorkManager, OkHttp, Coil, ZXing
Backend: FastAPI, Flask, Node.js, Express, WebSockets, REST APIs
AI / ML: TensorFlow, Keras, PyTorch, Transformers, EasyOCR, YOLOv8, SMPLer-X, Gemini API, Together AI
Databases / Cloud: AWS SageMaker, Firestore, Firebase Realtime Database, Firebase Storage, MySQL, SQLite
Architecture & Methodologies: Agile / Scrum, Kanban, MVC, MVVM, offline-first architecture, REST API design
Tools: Git, GitHub, Jira, Figma, Postman, Android Studio, VS Code, Linux/CLI, SSH

LANGUAGES

Spanish: Native
English: B2 – Professional working proficiency